



3832 - QUIESCENT BLACK HOLE X-RAY BINARIES IN THE MID-IR: RELATIVISTIC JET OR CIRCUMBINARY DISK?

Cycle: 2, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Elena Gallo (PI)	University of Michigan
Dr. Gabriele Cugno (CoI)	University of Michigan
Dr. Jon Matthew Miller (CoI)	University of Michigan
Prof. Rob Fender (CoI) (ESA Member)	University of Oxford
Dr. James Miller-Jones (CoI)	Curtin University
Dr. Richard M. Plotkin (CoI)	University of Nevada - Reno
Dr. David M. Russell (CoI)	New York University Abu Dhabi

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Imaging A0620	MIRI Imaging	(1) A0620-00
	2	Imaging J1118	MIRI Imaging	(2) XTEJ1118+480
	3	LRS A0620	MIRI Low Resolution Spectroscopy	(1) A0620-00
	4	LRS J1118	MIRI Low Resolution Spectroscopy	(2) XTEJ1118+480

ABSTRACT

JWST offers the opportunity to solve a long-standing puzzle about the nature of the mid-IR emission from quiescent Galactic black hole X-ray binaries, i.e., systems with X-ray luminosities well below 10^{-4} times their Eddington limit, L_{Edd} . Spitzer IRAC/MIPS observations of two prototypical such systems, 1A 0620-00 and XTE J1118+480 (10^{-9} and $10^{-9} L_{\text{Edd}}$, respectively) detected clear evidence for mid-IR excess

JWST Proposal 3832 (Created: Wednesday, January 24, 2024 at 9:00:12 AM Eastern Standard Time) - Overview

emission above the donor star photosphere spectra. The large error bars at 8 and 24 micron allow for radically different interpretations of this excess emission; either partially self-absorbed synchrotron emission from a relativistic jet, or, thermal emission from circumbinary material that reprocesses the light of the donor star. This GO program aims to leverage the superior sensitivity of JWST/MIRI to ascertain the physical nature of the mid-IR excess seen by Spitzer. If the circumbinary disk scenario proves to be correct, it may even account for the unusually large orbital decay rates of these systems. Either interpretation would have important consequences for our understating of how the jet power scales with accretion rate – a critical question across the black hole mass scale.

OBSERVING DESCRIPTION

MIRI LRS plus 4 band imaging observations of 2 low-luminosity Galactic X-ray binaries.

Proposal 3832 - Targets - QUIESCENT BLACK HOLE X-RAY BINARIES IN THE MID-IR: RELATIVISTIC JET OR CIRCUMBINARY D...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	A0620-00	RA: 06 22 44.5419 (95.6855912d) Dec: -00 20 44.37 (-.34566d) Equinox: J2000	Proper Motion RA: -2.926719926367103E-5 sec of time/yr Proper Motion Dec: -0.00513800002863718 arcsec/yr Parallax: 0" Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Black holes, X-ray binary stars] Extended=NO				
Fixed Targets	(2)	XTEJ1118+480	RA: 11 18 10.7651 (169.5448546d) Dec: +48 02 12.21 (48.03672d) Equinox: J2000	Proper Motion RA: -0.0018051185897870817 sec of time/yr Proper Motion Dec: -0.0066870000864582835 arcsec/yr Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Black holes, X-ray binary stars] Extended=NO				

Proposal 3832 - Observation 1 - QUIESCENT BLACK HOLE X-RAY BINARIES IN THE MID-IR: RELATIVISTIC JET OR CIRCUMBIN...

Observation	Proposal 3832, Observation 1: Imaging A0620										Wed Jan 24 14:00:12 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	A0620-00	RA: 06 22 44.5419 (95.6855912d) Dec: -00 20 44.37 (-.34566d) Equinox: J2000			Proper Motion RA: -2.926719926367103E-5 sec of time/yr Proper Motion Dec: -0.00513800002863718 arcsec/yr Parallax: 0" Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Black holes, X-ray binary stars] Extended=NO										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		5	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F1500W	FASTR1	6	1	1	Dither 1	4	4	66.601	
	2	F1800W	FASTR1	6	1	1	Dither 1	4	4	66.601	
	3	F2100W	FASTR1	6	1	1	Dither 1	4	4	66.601	
	4	F2550W	FASTR1	10	7	1	Dither 1	4	28	843.612	

Proposal 3832 - Observation 2 - QUIESCENT BLACK HOLE X-RAY BINARIES IN THE MID-IR: RELATIVISTIC JET OR CIRCUMBIN...

Observation	Proposal 3832, Observation 2: Imaging J1118										Wed Jan 24 14:00:12 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	XTEJ1118+480	RA: 11 18 10.7651 (169.5448546d) Dec: +48 02 12.21 (48.03672d) Equinox: J2000			Proper Motion RA: -0.0018051185897870817 sec of time/yr Proper Motion Dec: -0.0066870000864582835 arcsec/yr Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Black holes, X-ray binary stars] Extended=NO										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	3		5	1			LARGE	
	2	CYCLING	1	12						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F1500W	FASTR1	8	1	1	Dither 1	3	3	66.601	11
	2	F1800W	FASTR1	32	1	1	Dither 1	3	3	266.404	11
	3	F2100W	FASTR1	28	1	1	Dither 2	12	12	932.413	11
	4	F2550W	FASTR1	15	4	1	Dither 2	12	48	2097.93	11

Proposal 3832 - Observation 3 - QUIESCENT BLACK HOLE X-RAY BINARIES IN THE MID-IR: RELATIVISTIC JET OR CIRCUMBIN...

Observation	Proposal 3832, Observation 3: LRS A0620										Wed Jan 24 14:00:12 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Low Resolution Spectroscopy										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	A0620-00	RA: 06 22 44.5419 (95.6855912d) Dec: -00 20 44.37 (-.34566d) Equinox: J2000			Proper Motion RA: -2.926719926367103E-5 sec of time/yr Proper Motion Dec: -0.00513800002863718 arcsec/yr Parallax: 0" Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Black holes, X-ray binary stars] Extended=NO										
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID		
	1	SAME	F560W	FAST	6	1	1	16.65	12		
Template	Subarray					Obtain Verification Image?					
	FULL					true					
Dithers	#	Dither Type	No. Spectral Steps			Spectral Step Offset		No. Spatial Steps		Spatial Step Offset	
	1	ALONG SLIT NOD									
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter	
	1	FASTR1	6	1	1	1	1	16.65		F560W	

Proposal 3832 - Observation 3 - QUIESCENT BLACK HOLE X-RAY BINARIES IN THE MID-IR: RELATIVISTIC JET OR CIRCUMBIN...

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	23	5	10	1	2	660.46	9

Proposal 3832 - Observation 4 - QUIESCENT BLACK HOLE X-RAY BINARIES IN THE MID-IR: RELATIVISTIC JET OR CIRCUMBIN...

Observation	Proposal 3832, Observation 4: LRS J1118 Wed Jan 24 14:00:12 GMT 2024								
	Diagnostic Status: Warning Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(2)	XTEJ1118+480	RA: 11 18 10.7651 (169.5448546d) Dec: +48 02 12.21 (48.03672d) Equinox: J2000		Proper Motion RA: -0.0018051185897870817 sec of time/yr Proper Motion Dec: -0.0066870000864582835 arcsec/yr Epoch of Position: 2015.5		<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Star</i> <i>Description=[Black holes, X-ray binary stars]</i> <i>Extended=NO</i>		
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SAME	F560W	FAST	6	1	1	16.65	12
Template	Subarray				Obtain Verification Image?				
	FULL				true				
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps	
	1	ALONG SLIT NOD							
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID
	1	FASTR1	6	1	1	1	1	16.65	F560W

Proposal 3832 - Observation 4 - QUIESCENT BLACK HOLE X-RAY BINARIES IN THE MID-IR: RELATIVISTIC JET OR CIRCUMBIN...

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	45	20	40	1	2	5100.524	9